Ryann Sim Wei Jian

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Research Interests

My primary research interest is in studying multiagent systems through the lens of game theory, machine learning and optimization. In particular, I seek to understand the algorithmic properties of discrete-time learning algorithms in various game settings, while also drawing connections to continuous-time dynamics which are prevalent in evolutionary game theory. My recent work has also explored game-theoretic problems at the interface between complexity theory, polynomial optimization and games of incomplete information. As a secondary research interest, I am also broadly interested in applying learning-in-games and social choice theory ideas to study the behavior of human-AI interactions, and to develop fairer and more stable AI systems.

Academic Experience

School of Computing, National University of Singapore *Singapore*

Postdoctoral Research Fellow *Oct. 2025 - Oct. 2026*

- Supervised by Chun Kai Ling.

Singapore University of Technology and Design *Singapore*

Postdoctoral Research Fellow *Aug. 2024 - Sep. 2025*

- Supervised by Antonios Varvitsiotis and Georgios Piliouras. Research focus on learning in quantum games and polynomial optimization techniques for learning in extensive-form games.

Education

Singapore University of Technology and Design *Singapore*

Ph.D., Engineering Systems and Design *Sep. 2019 - May 2024*

- Awarded SUTD President's Graduate Fellowship
- Supervised by Georgios Piliouras
- Thesis Committee: Rakesh Nagi, Antonios Varvitsiotis, Dario Poletti

Singapore University of Technology and Design *Singapore*

B.Eng., Engineering Systems and Design (Summa Cum Laude) *May 2016 - Aug. 2019*

- Awarded Wilmar Scholarship
- Focus Track: Operations Research

Journal Publications

1. Wayne Lin, Georgios Piliouras, *Ryann Sim*, and Antonios Varvitsiotis. **Learning in Quantum Common-Interest Games and the Separability Problem**. *Quantum* 9 (2025): 1689.
2. Wayne Lin, Georgios Piliouras, *Ryann Sim*, and Antonios Varvitsiotis. **No-Regret Learning and Equilibrium Computation in Quantum Games**. *Quantum* 8 (2024): 1569.

Conference Publications

1. Rui Zheng, *Ryann Sim*, and Antonios Varvitsiotis. **Solving Imperfect-Recall Games via Sum-of-Squares Optimization**. *International Conference on Machine Learning*, 2026.
2. Vincent Leon, Iosif Sakos, *Ryann Sim*, and Antonios Varvitsiotis. **Certifying Concavity and Monotonicity in Games via Sum-of-Squares Hierarchies**. *Advances in Neural Information Processing Systems*, 38, 2025.
3. John Lazarsfeld, Georgios Piliouras, *Ryann Sim*, and Stratis Skoulakis. **Optimism Without Regularization: Constant Regret in Zero-Sum Games**. *Advances in Neural Information Processing Systems*, 38, 2025.
4. John Lazarsfeld, Georgios Piliouras, *Ryann Sim*, and Andre Wibisono. **Fast and Furious Symmetric Learning in Zero-Sum Games: Gradient Descent as Fictitious Play**. *Conference on Learning Theory*, 2025.

5. Volkan Cevher, Georgios Piliouras, *Ryann Sim*, and Stratis Skoulakis. **Min-Max Optimization Made Simple: Approximating the Proximal Point Method via Contraction Maps**. *SIAM Symposium on Simplicity in Algorithms*, 2023.
6. Rahul Jain, Georgios Piliouras, and *Ryann Sim*. **Matrix Multiplicative Weights Updates in Quantum Zero-Sum Games: Conservation Laws & Recurrence**. *Advances in Neural Information Processing Systems*, 35, 2022.
7. Georgios Piliouras, *Ryann Sim*, and Stratis Skoulakis. **Beyond Time-Average Convergence: Near-Optimal Uncoupled Online Learning via Clairvoyant Multiplicative Weights Update**. *Advances in Neural Information Processing Systems*, 35, 2022.
8. Georgios Piliouras, Lillian J Ratliff, *Ryann Sim*, and Stratis Skoulakis. **Fast Convergence of Optimistic Gradient Ascent in Network Zero-Sum Extensive Form Games**. In *International Symposium on Algorithmic Game Theory*. Springer, 2022.
9. Tanner Fiez, *Ryann Sim*, Stratis Skoulakis, Georgios Piliouras, and Lillian Ratliff. **Online Learning in Periodic Zero-Sum Games**. *Advances in Neural Information Processing Systems*, 34, 2021.
10. Stratis Skoulakis, Tanner Fiez, *Ryann Sim*, Georgios Piliouras, and Lillian Ratliff. **Evolutionary Game Theory Squared: Evolving Agents in Endogenously Evolving Zero-Sum Games**. In *AAAI Conference on Artificial Intelligence*, 2021.

Preprints

1. Rui Zheng, *Ryann Sim*, and Antonios Varvitsiotis. **Welfare Optimization over Semi-Algebraic Equilibrium Sets**. Preprint, 2026.
2. Tom Schwarz, Jiaru Li, *Ryann Sim*, and Chun Kai Ling. **Computing Equilibria in Games with Stochastic Action Sets**. Preprint, 2026.
3. Tom Schwarz, Jiaru Li, *Ryann Sim*, and Chun Kai Ling. **Equilibrium Computation in Extensive-Form Games with Stochastic Action Sets**. Preprint, 2026.

Teaching and Mentoring

Student Mentoring

- Jiaru Li, Master's student @ National University of Singapore — *May 2026 - Oct 2026*
- Tom Schwarz, PhD student @ National University of Singapore — *Oct 2025 - Oct 2026*
- Rui Zheng, PhD student @ Singapore University of Technology and Design — *Sep 2024 - Sep 2025*

Engineering Systems and Design Pillar, SUTD, Singapore

- Co-Instructor, 40.002 Optimization (Instructor Rating: 4.56/5) — *Spring 2025*
- Graduate Teaching Assistant, 40.012 Manufacturing and Service Operations — *Spring 2021*
- Graduate Teaching Assistant, 40.015 Simulation Modeling and Analysis — *Fall 2020*
- Undergraduate Teaching Assistant, 40.014 Systems Architecture — *Spring 2019*

Faculty of Science and Math, SUTD, Singapore

- Undergraduate Teaching Assistant, 10.004 Advanced Mathematics — *Fall 2018*
- Instructor, Freshmore Learning Hub — *Fall 2018*
- Undergraduate Teaching Assistant, 10.007 Systems World — *Spring 2018*
- Undergraduate Teaching Assistant, 3.007 Introduction to Design — *Fall 2017*
- Undergraduate Teaching Assistant, 10.001 Mathematics Bootcamp — *Summer 2017*

Independent Activity Period, SUTD, Singapore

- Instructor, Introduction to LaTeX — *Jan. 2019*
- Instructor, Introduction to HTML and CSS — *Jan. 2017*

Honors & Awards

- **ESD Outstanding Graduate Teaching Assistant Award**, SUTD *2025*
- **NeurIPS Top Reviewer** *2025*
- **ESD Outstanding Graduate Teaching Assistant Award**, SUTD *2021*
- **President's Graduate Fellowship**, SUTD *2019*
- **ESD Outstanding Undergraduate Teaching Assistant Award**, SUTD *2019*
- **Meritorious Winner**, COMAP Interdisciplinary Contest in Modeling (ICM) Problem D *2019*
- **Math II Honors Class (Linear Algebra)**, Faculty of Science and Math, SUTD *2016*
- **Wilmar Scholarship**, Wilmar International *2016*

Talks & Presentations

- Invited Talk: 'Advancing Dynamics and Equilibrium Computation Beyond Classical Learning in Games'. *Singapore University of Technology and Design, Singapore, Jun. 2026*
- Invited Talk: 'Advancing Dynamics and Equilibrium Computation Beyond Classical Learning in Games'. *Shanghai University of Finance and Economics, China, Feb. 2026*
- Invited Talk: 'No-Regret Learning and Equilbrium Computation in Quantum Games'. *INFORMS International, Singapore, Jul. 2025*
- Invited Talk: 'Certifying Concavity and Monotonicity in Games via Sum-of-Squares Hierarchies'. *King's College London, UK, Jul. 2025*
- Invited Talk: 'On Un(der)-Regularized Learning in Symmetric Zero-Sum Games'. *University of Adelaide, Australia, Feb. 2025*
- Poster Presentation: 'Matrix Multiplicative Weights Update in Quantum Zero-Sum Games: Conservation Laws and Recurrence'. *NeurIPS, New Orleans, USA, Dec. 2022*
- Poster Presentation: 'Beyond Time-Average Convergence: Near-Optimal Uncoupled Online Learning via Clairvoyant Multiplicative Weights Update'. *NeurIPS, New Orleans, USA, Dec. 2022*
- Oral Presentation: 'Fast Convergence of Optimistic Gradient Ascent in Network Zero-Sum Extensive Form Games'. *SAGT, Colchester, UK, Sep. 2022*
- Virtual Talk: 'Online Learning in Periodic Zero-Sum Games'. *NeurIPS (Online), Dec. 2021*
- Virtual Poster Presentation: 'Evolutionary Game Theory Squared: Evolving Agents in Endogenously Evolving Zero-Sum Games'. *AAAI (Online), Feb. 2021*
- Invited Talk: 'Not-So-Simple Linear Regression'. *PyData Kuala Lumpur, Jul. 2020*

Relevant Coursework

Linear Optimization, Convex Optimization, Stochastic Modeling, Statistics, Optimization for Machine Learning, Algorithmic Game Theory, Algorithms, Networked Life, Statistical and Machine Learning, Game Theory, Advanced Optimization (Scheduling and Queueing Theory).

Academic Service

- Area Chair: ICLR (2027)
- Program Committee: AAAI (2025-2027), EC (2026), ALT (2027).
- Journal reviewer: Royal Society Proceedings A (2025), TMLR (2026), IEEE TACON (2026).
- Conference reviewer: NeurIPS (2021,2023,2025-2026), ICML (2022,2026), ICLR (2024-2026), COLT (2026), AAAI (2020-2022), AAMAS (2027).
- Sub-reviewer: WINE (2020), STOC (2021), AISTATS (2020-2021), ICDCS (2022), AAMAS (2023).
- SUTD Graduate Student Association (Vice-President, AY20/21).

Additional Experience

Shanghai University of Finance and Economics *Shanghai, China*

Research Intern *Feb. 2023 - May 2023*

- Hosted by Wang Xiao in ITCS Department, focusing on studying the dynamical behavior of alternating online algorithms.

British Petroleum Singapore *Singapore*

Core Data Strategy Intern *May 2018 - Aug. 2018*

Languages and Skills

Programming	Proficient in Python, R, LaTeX, SQL, HTML and CSS
	Intermediate in Julia, JavaScript, Mathematica, MATLAB, AMPL, VBA and PyTorch
Languages	Fluent in English, Mandarin and Malay (written and spoken)
	Intermediate in Cantonese and Hokkien

References

References available upon request.